Motion with Constant Acceleration

When an object moves with constant acceleration a, its velocity changes by the same amount every second. If it starts with initial velocity u, after time t its velocity is v = u + a·t.

The distance travelled in that time is s = u·t + ½·a·t². Combining the two equations to remove t gives v² = u² + 2·a·s, which is useful when the time is not known.

Worked example

A car starts from rest (u = 0 m/s) and accelerates at a = 2 m/s² for t = 5 s. Its final velocity is v = 0 + 2 × 5 = 10 m/s, and it travels s = ½ × 2 × 5² = 25 m.

Summary table

Symbol	Meaning	Unit
u	initial velocity	m/s
v	final velocity	m/s
a	acceleration	m/s²
t	time	s
s	displacement	m